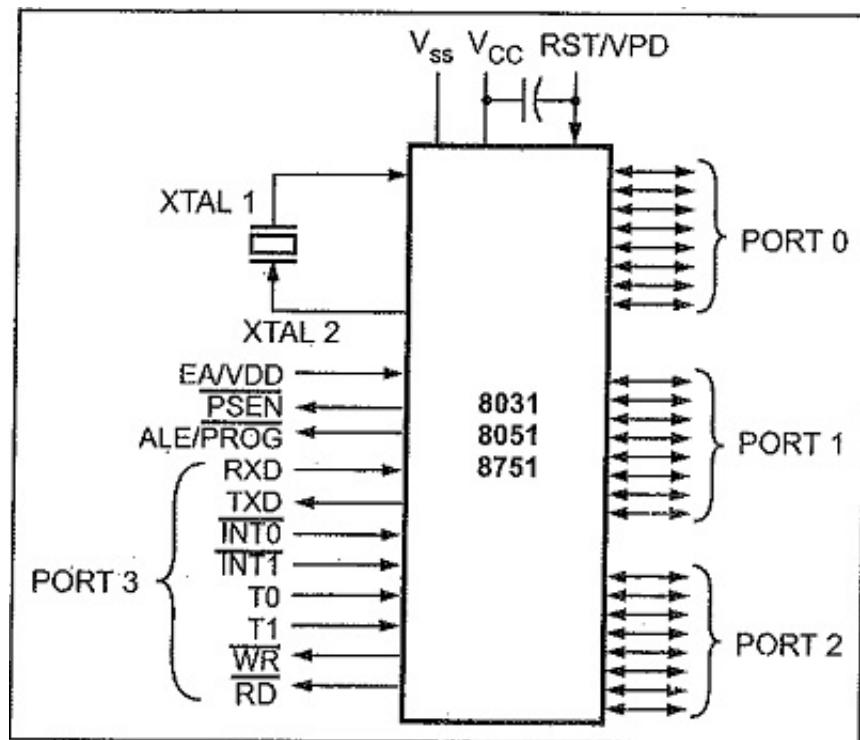


Embedded Computing Principles

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8051 Microcontroller – Summary



The 8051 is an 8-bit microcontroller originally developed by Intel, widely used in embedded systems. The diagram shows three related chips from the same family: **8031**, **8051**, and **8751** (they differ mainly in on-chip memory type — the 8031 has no internal ROM, the 8051 has masked ROM, and the 8751 has EPROM).

Key Pins/Features from the Diagram

Power & Clock:

- **V_{ss} / V_{CC}** – Ground and supply voltage pins
- **XTAL1 / XTAL2** – Connections for an external crystal oscillator that drives the internal clock circuitry
- **RST/VPD** – Reset pin; also used for power-down mode backup voltage

Control Signals:

- **EA/VDD** (External Access) – Determines whether the controller executes from internal or external program memory
- **PSEN** (Program Store Enable) – Read strobe for external program memory
- **ALE/PROG** (Address Latch Enable) – Latches the lower byte of the address during access to external memory

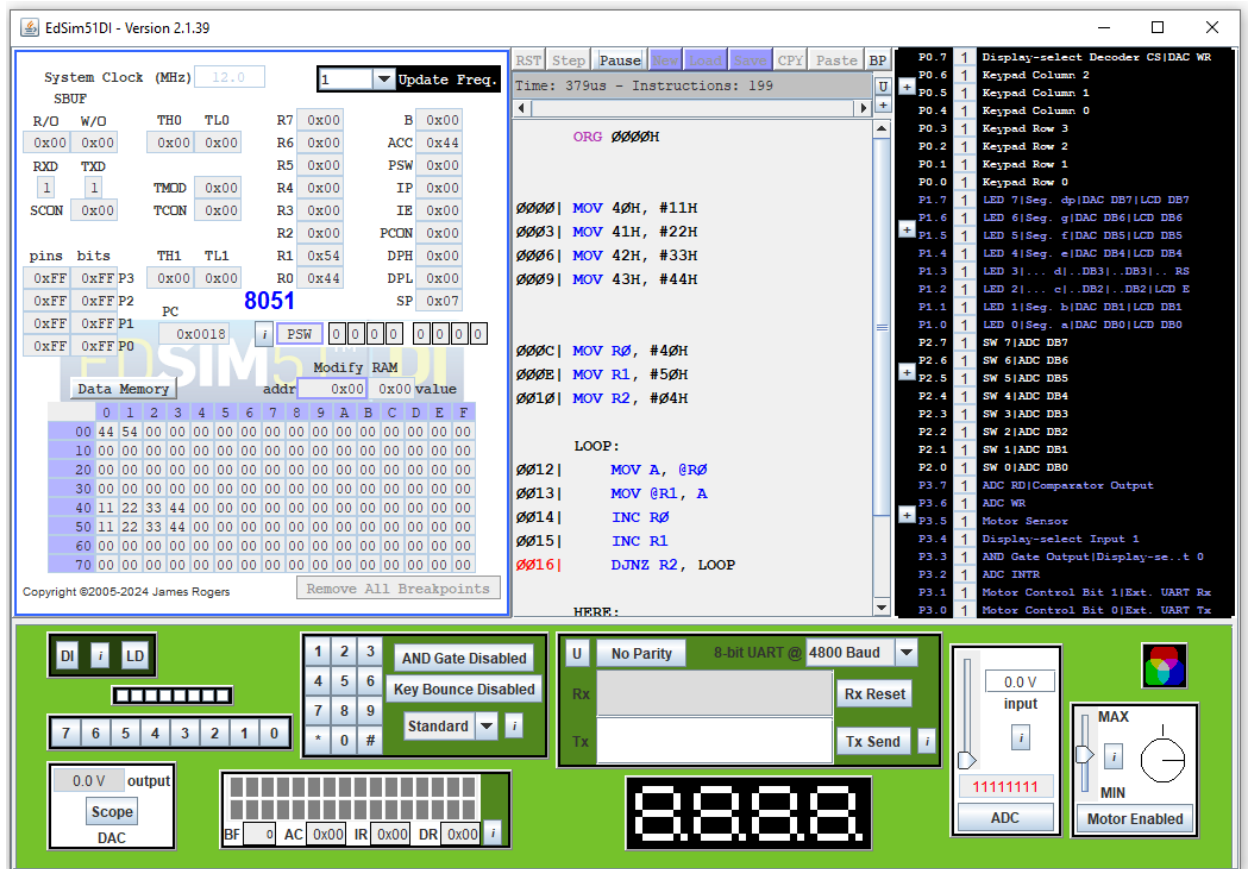
I/O Ports (each 8-bit, bidirectional):

- **Port 0** – General-purpose I/O; also serves as multiplexed address/data bus for external memory
- **Port 1** – Dedicated general-purpose I/O port
- **Port 2** – General I/O; also provides the higher-order address byte for external memory
- **Port 3** – General I/O with important **alternate functions**:
 - RXD, TXD – Serial communication (UART)
 - INT0, INT1 – External interrupts
 - T0, T1 – Timer/counter external inputs
 - WR, RD – Write and read control signals for external data memory

Core Architectural Highlights (general 8051 knowledge)

- 8-bit CPU with Boolean (bit-level) processing capability
- 4 KB on-chip ROM (in 8051 variant), 128 bytes internal RAM
- Two 16-bit timers/counters
- Full-duplex serial UART
- 4 I/O ports (32 I/O lines total)
- 5 interrupt sources (2 external, 3 internal)
- Address bus capable of accessing 64 KB external memory (both program and data)

Q. Write a program to copy 4 data elements from 1 memory block (40H to 45H) to another memory block.

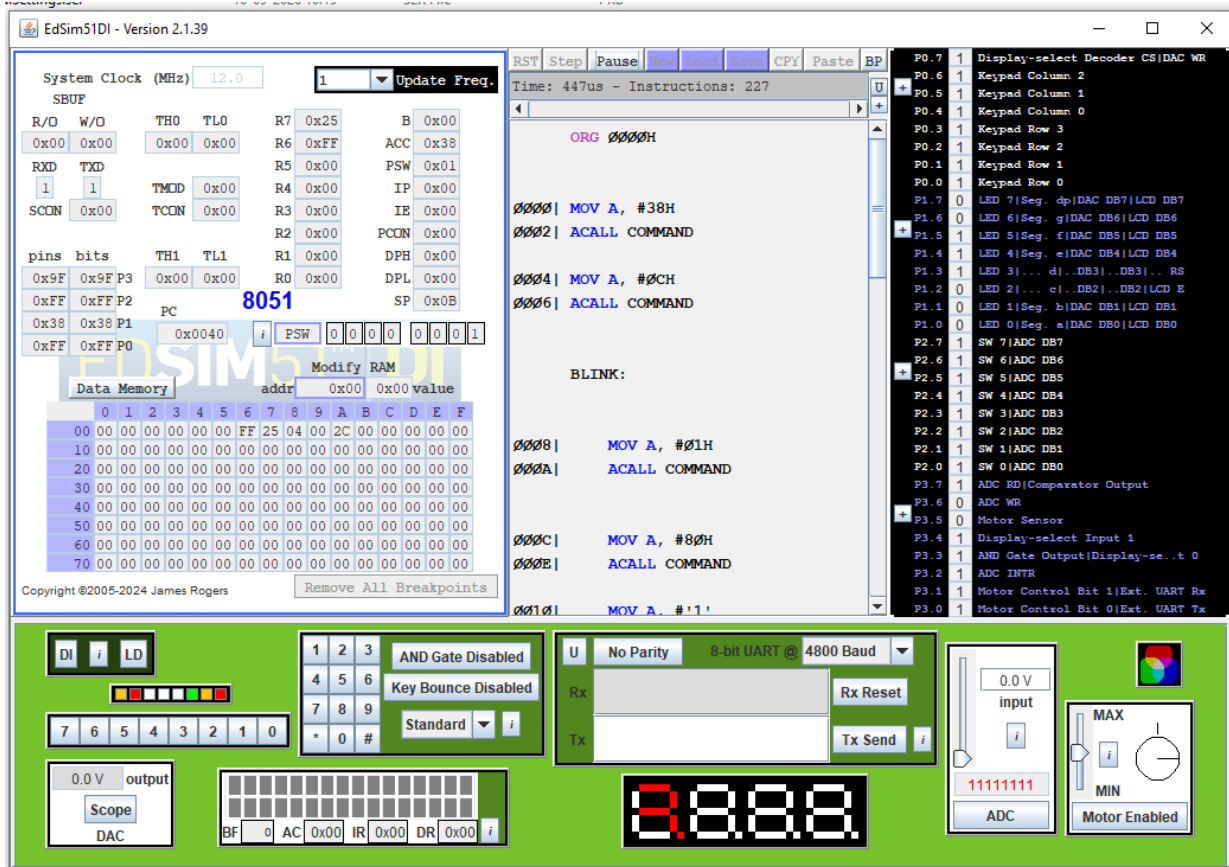


CODE:

```

ORG 0000H
MOV 40H, #11H
MOV 41H, #22H
MOV 42H, #33H
MOV 43H, #44H
MOV R0, #40H
MOV R1, #50H
MOV R2, #04H
LOOP:
    MOV A, @R0
    MOV @R1, A
    INC R0
    INC R1
    DJNZ R2, LOOP
HERE:
    SJMP HERE
END
  
```

Q. Write a program to blink the lcd display (2 digit of lcd display) in edsim51



CODE:

ORG 0000H

MOV A, #38H

ACALL COMMAND

MOV A, #0CH

ACALL COMMAND

BLINK:

MOV A, #01H

ACALL COMMAND

MOV A, #80H

ACALL COMMAND

MOV A, #'1'

ACALL DATA

MOV A, #'2'

ACALL DATA

ACALL DELAY

MOV A, #01H

ACALL COMMAND

ACALL DELAY

SJMP BLINK

COMMAND:

MOV P1, A

CLR P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DATA:

MOV P1, A

SETB P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DELAY:

MOV R6, #255

D1:

MOV R7, #255

D2:

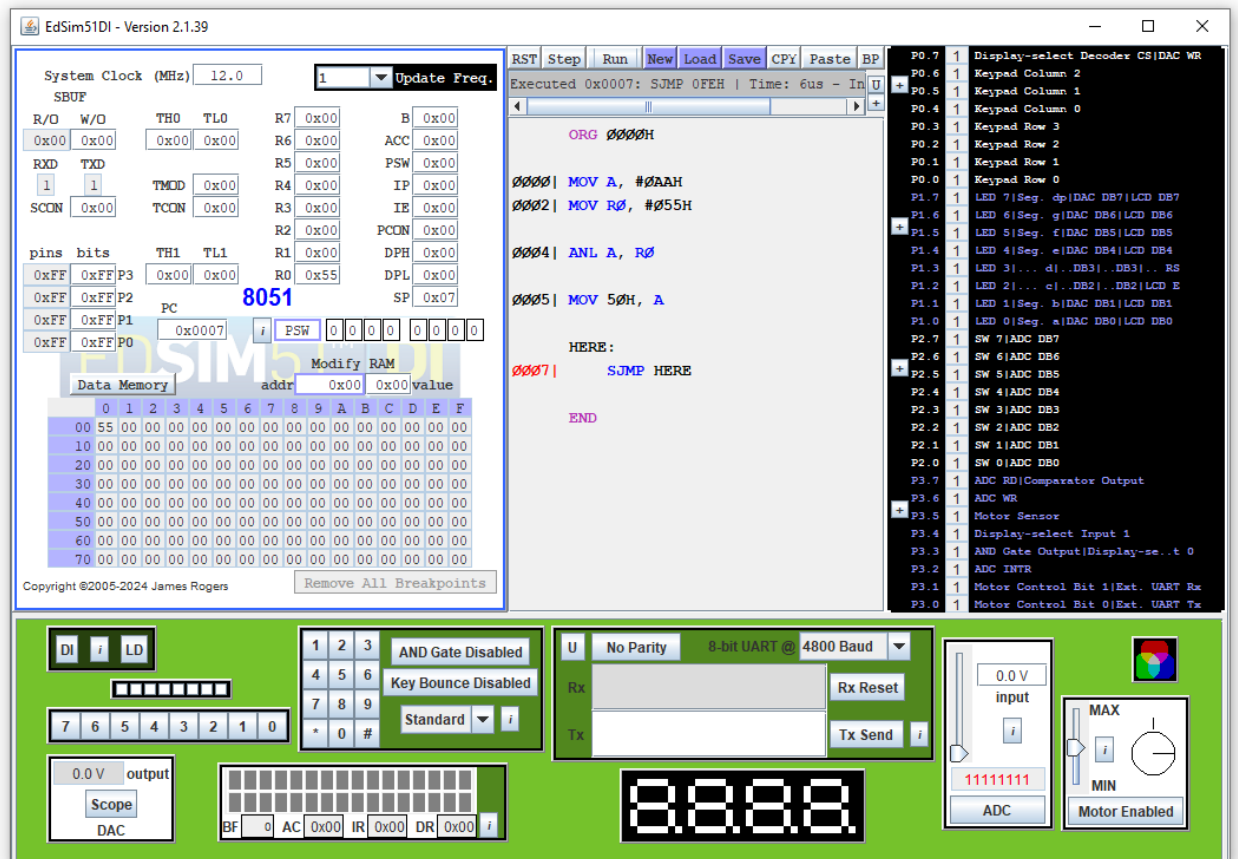
DJNZ R7, D2

DJNZ R6, D1

RET

END

Q. Write a program to perform AND operation



CODE:

ORG 0000H

MOV A, #0AAH

MOV R0, #055H

ANL A, R0

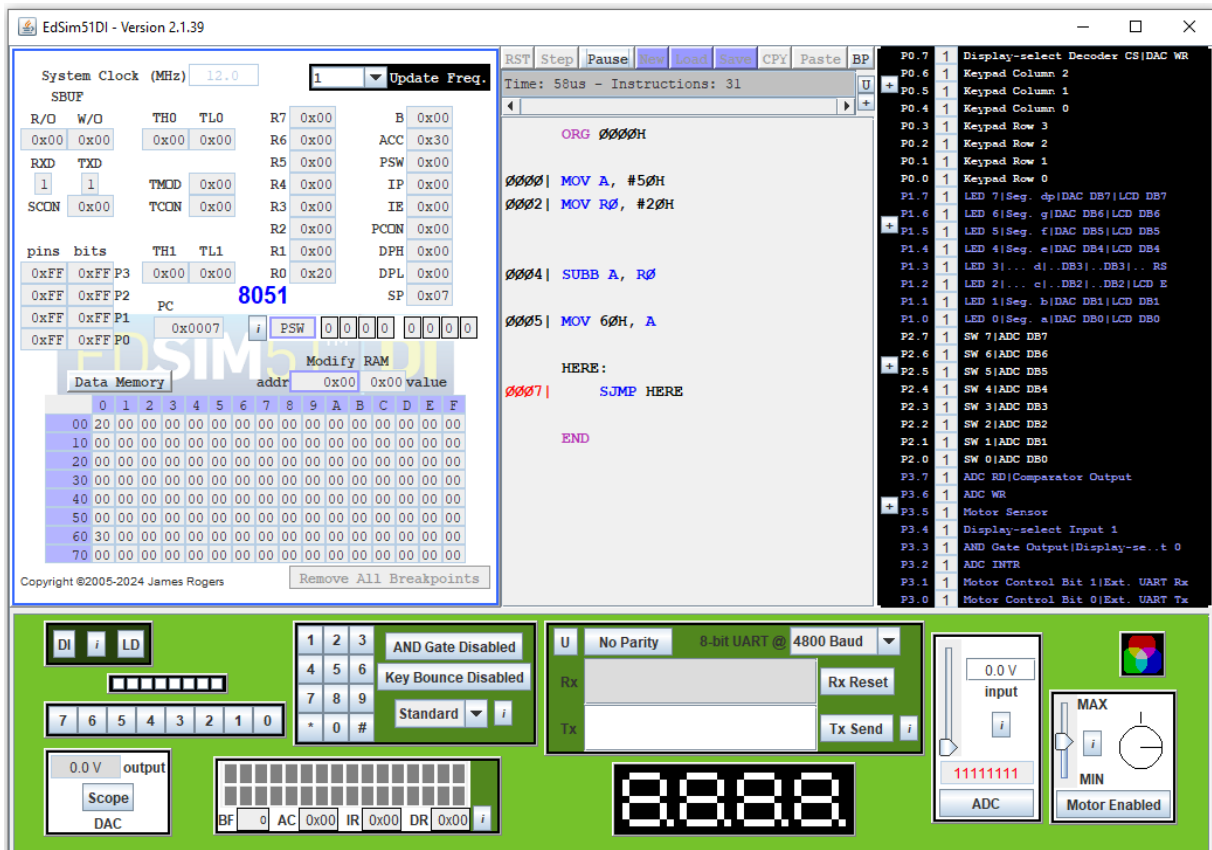
MOV 50H, A

HERE:

SJMP HERE

END

Q. Write a program to perform subtraction 20h FROM 50H



CODE:

ORG 0000H

MOV A, #50H

MOV R0, #20H

SUBB A, R0

MOV 60H, A

HERE:

SJMP HERE

END